Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

Abstract

Dealers set limits on the largest absolute position reached before a review horizon ends. We compare conditional quantile forecasts with historical rules using an hourly FX record, its daily aggregates and weekly dealer positions. Simple historical rules are hard to beat. Long-history FX regression lowers matched loss by 14.8% and 16.6% across periods. Delayed calibration retains the later gain (15.0%), with 329 versus 328 breaches. Rankings against a shorter rolling rule vary by period. Expanded regularization leaves regression behind in 34 of 48 weekly panel comparisons, and historical rules win most daily comparisons. Stationary controlled simulations already produce large short-history losses. In newly calibrated simulations at history 52, the full regression's median loss increase across series is 116.2% under constant parameters and 121.4% under parameter switching. The mechanism behind the observed low-frequency losses remains unidentified. Peak-limit validation should assess accuracy, breaches and reference stability together.

Keywords: Peak net positions; Quantile forecasting; Historical benchmarks; Forecast comparison; Calibration; Dealer positions.

1 Introduction

Dealer position limits concern the largest exposure reached before a review horizon ends. A position can finish within its limit after crossing it earlier (Figure 1). We compare historical and conditional forecasts of this sampled absolute-position peak.

Historical headroom combines the current position with completed past peak behavior. Conditional models estimate additional relationships from finite historical windows. Matched histories and update schedules show whether the added conditioning improves the forecast.

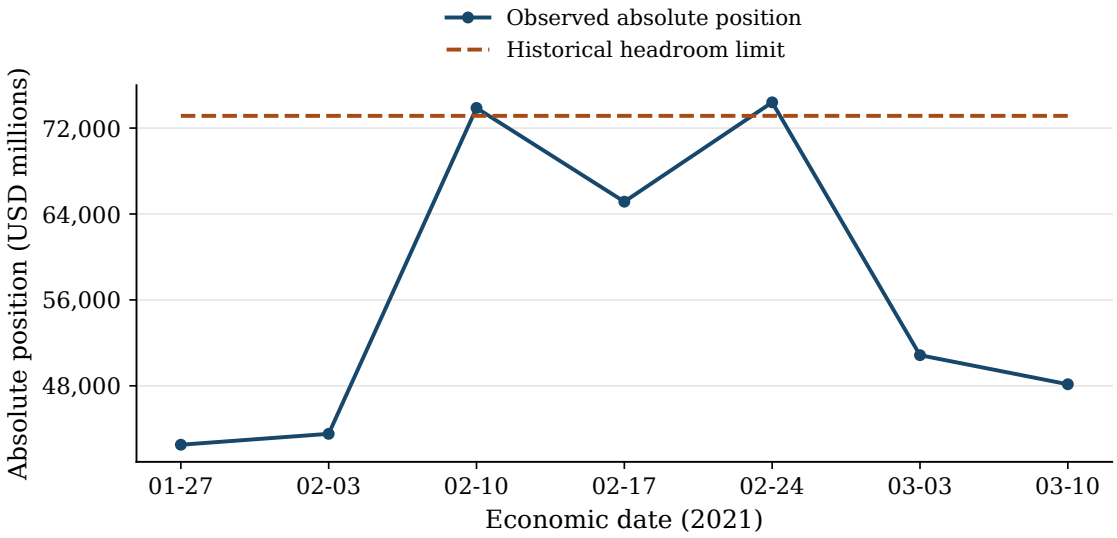


Figure 1: A selected Treasury-bill path whose endpoint returns within the historical report after an intervening breach. The rule chooses the earliest eligible original evaluation window. Endpoint containment can conceal a peak breach.

Our comparisons cover hourly FX positions, their daily aggregates and weekly U.S. dealer positions. Each series retains its source units and evaluation grid. Two comparisons from the

original execution protocol anchor the analysis. They use a shorter rolling FX reference and matched Treasury-bill headroom, respectively.

The paper makes three empirical points.

First, matched FX gains recur across periods, and delayed calibration retains the later gain. Regression lowers matched loss by 14.8% and 16.6% (Table 3). Calibration retains a 15.0% gain with 329 versus 328 breaches. The earlier ranking reverses against the shorter rolling reference. Before calibration, the matched gain accompanies 376 versus 360 breaches.

Second, historical rules usually prevail in the weekly and same-record daily comparisons. Expanded regression still loses 34 of 48 weekly panel comparisons (Table 4). Historical headroom wins seven of eight daily comparisons. Wider penalties and conservative validation leave substantial weekly losses, including a worst-cell increase of 181.4%.

Third, stationary processes can produce large short-history regression losses. The controlled stationary design already raises 33-predictor loss by 113.5–135.5% at history 52. The newly fitted constant control raises loss by 116.2%, compared with 121.4% under parameter switching. Section 7.4 compares the designs and the incomplete longer-history results. The proposed instability explanation remains unidentified.

Section 3 defines the target and records two elementary properties used later. The classical endpoint-to-peak bound supplies a separate benchmark under its stated assumptions. Subsequent sections compare matched forecasts, their stability and their implications for monitoring limits.

2 Related literature

Wisniewski and colleagues forecast signed next-hour position intervals using historical states, calendar information and local prediction methods (Wisniewski et al., 2020; Wisniewski, 2024). Their chronological training, calibration and test design reports scores and coverage together. The present peak response uses a complete future path segment and the current magnitude, producing different labels and limits.

Path-risk forecasts address within-period extremes. MaxVaR studies the worst cumulative portfolio return under fixed holdings and diffusion assumptions (Boudoukh et al., 2004). The payment-liquidity thesis forecasts conditional maximum deficits from confidential transactions (van den Hoogen, 2017). Settlement studies examine tail liquidity needs and collateral requirements (Petrunia et al., 2019; Arjani et al., 2020). Duffie and colleagues study dealer capacity and Treasury-market functioning, including PnL quantile forecasts (Duffie et al., 2023). These responses differ from the sampled absolute-position peak.

Regression quantiles target asymmetric forecast errors directly (Koenker & Bassett, 1978). Autoregressive quantiles and quantile forests supply other conditioning structures (Engle & Manganelli, 2004; Meinshausen, 2006). Consistent scores and elementary-score representations permit comparison across evaluation preferences (Gneiting, 2011; Ehm et al., 2016). Table 1 locates the present peak response, completed labels and matched historical controls.

The empirical contribution concerns matched peak forecasts and the stability of their rankings. The forecasting and scoring methods are established. Tables 3–5 connect accuracy to breaches; Section 7.4 compares distinct stationary and switching designs.

Table 1: Closest literature and the peak-position task. A dash marks a dimension unverified in the consulted passage. Established methods support the comparison of a common peak response.

Study	Response	Data unit	Peak floor	Label timing	Evaluation	Source scope
Wisniewski et al. (2020)	Signed endpoint interval	One FX market maker	—	Ordered train/ calibrate/ test	Scores and hit tests	One source family
Boudoukh et al. (2004)	Worst portfolio return	Stylized fixed portfolio	Different initial-value anchor	Analytic diffusion	Tail ratios and monitoring	Parameter cases
van den Hoogen (2017)	Cumulative payment-deficit peak	Confidential bank payments	Absolute-position floor —	Path simulation	EVT methodology; results partly confidential	Bank payment task
Duffie et al. (2023)	Capacity, illiquidity, PnL quantiles	Dealer/ market sources	—	Expanding PnL fits	Capacity and PnL forecasts	Several sources
Engle and Manganelli (2004)	Return quantile	Asset returns	—	Lagged predictors	Quantile scores and DQ	Several returns
Ehm et al. (2016)	Quantile/ expectile comparison	Multiple forecast examples	—	Original example designs	Elementary-score curves	Multiple tasks
Meinshausen (2006)	Conditional response quantile	Regression examples	—	Original regression splits	Conditional distributions	Several datasets
This study	Sampled absolute-position peak	One FX record; weekly totals	Observed current magnitude	Completed forward labels	Scores, breaches, coherence	Related aggregate tasks

The payment-thesis reading is partial. Dashes mark unverified dimensions in the consulted passages; shared source lineage leaves input-byte identity unverified.

3 The forecast target

The target is the largest sampled absolute position from issue time through the stated horizon. Let N_t denote signed position and $\mathcal{F}_t$ the issue-time information. The peak $Y_{t,H}$, observed floor B_t and nonnegative headroom $Z_{t,H}$ satisfy the following identities.

$$Y_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad B_t = |N_t|, \qquad Z_{t,H} = Y_{t,H} - B_t \ge 0.$$

A feasible limit covers the current observed exposure. Figure 1 shows an endpoint inside the historical report after an earlier breach. We score reports with pinball loss and count a strict breach when the peak exceeds the limit.

$$\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+, \qquad I_{t,H} = \mathbf{1}\{Y_{t,H} > C_{t,H}\}.$$

Proposition 1(known-floor projection). Suppose $Y \ge B$ almost surely and the issue information contains B. Projecting any forecast c to $\max\{B, c\}$ weakly improves pinball loss for every possible outcome, with an exact gain below the floor. If q is the true conditional lower inverse quantile at level $1-\alpha$, its strict exceedance probability is at most α. At an atom of the response distribution, that probability can be smaller than the nominal tail probability.

$$\rho_\tau(Y - c) - \rho_\tau(Y - \max\{B, c\}) = \tau(B - c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \le \alpha.$$

Proposition 2(classical endpoint-to-peak bound). Let future increments be independent and individually symmetric about zero, with partial sums S_h. Fix a starting position x, set $B = |x|$, and let q be the population $(1-\alpha/2)$ quantile of $|S_H|$. The triangle inequality and the classical scalar Lévy maximal inequality then imply the following bound, including strict inequalities at atoms (Sussman, 2023). Neither identical increment distributions nor finite increment moments are required for this population statement.

$$\Pr\left(\max_{0 \le h \le H} |x + S_h| > B + q\right) \le \Pr\left(\max_h |S_h| > q\right) \le 2\Pr(|S_H| > q) \le \alpha.$$

The Supplement proves both statements and compares population quantiles with their empirical plug-ins. A conditional application requires independence, symmetry and the stated endpoint quantile under the law given issue information. The separate bound simulations distinguish mean-reverting levels from serially correlated increments.

Separately fitted horizon forecasts can violate the ordering of true peak quantiles. Short-history FX regression crosses across horizons on 640 evaluation origins. Label nesting, floor coverage and strict-breach definitions provide additional checks on the issued reports.

4 Data

The FX application uses the hourly Net Position Time Series Dataset, reporting market-maker client-flow positions (Lindsay, 2020; Wisniewski et al., 2020). It contains 18,576 records in source USD units. The later, earlier and full-prefix comparisons use distinct chronological issue grids and the same sampled-peak definition.

The weekly data are New York Fed Primary Dealer Statistics (Federal Reserve Bank of New York, n.d.). Reporting definitions follow the government-securities dealer framework and position-accounting instructions (Board of Governors of the Federal Reserve System, 2013, 2024). The

Treasury-bill series has 706 economic-date observations in USD millions. The values are fair-value positions aggregated as long-minus-short totals. Additional categories pass metadata and calendar checks and use complete weekly grids, with separate fits by category.

Daily aggregation takes the last recorded position on each nonempty source-naive calendar day. It retains partial boundary days, yielding 930 days and 501 common evaluation issues. Reports use the completed day's end as their issue clock. The resulting target is a peak over subsequent daily records.

Every method within a comparison uses identical peak labels, current floors and issue origins. Source units and histories remain distinct across series. The same-record aggregation holds dealer identity fixed while changing the measurement clock.

5 Forecasting methods

Forecasts target the conditional quantile at $\tau = 19/20$. The horizons $H \in \{6, 24\}$ count subsequent source records, and each peak includes the issue position. Calendar gaps extend an FX horizon's elapsed span; weekly and daily steps represent different review intervals. The nineteen-to-one error-weight preference defines the evaluation score.

Historical headroom uses the corrected empirical rank of completed headrooms and adds its report to the current magnitude. Translated paths reanchor historical signed paths, while historical-peak quantiles are projected to the observed floor. The endpoint plug-in uses an absolute-displacement quantile; Table 2 gives the executed menus and schedules.

Ridge quantile regression links mature headroom to issue-known position states and calendar variables. The larger set contains 33 predictors plus an unpenalized intercept. The four-predictor variant uses current magnitude, signed level and two increment RMS measures. All standardization uses mature training observations. Expanded-menu regression changes only the penalty grid.

Local quantiles use neighboring historical states, with an increment-scaled variant. Filtered paths simulate positions from GARCH-standardized empirical innovations drawn independently with replacement (Hull & White, 1998; Barone-Adesi et al., 1999). Quantile forests use leaf distributions; tail methods fit volatility-scaled headroom (Meinshausen, 2006; McNeil & Frey, 2000). The Supplement lists complete results and numerical or structural failures.

Further comparisons apply delayed calibration, fixed combinations and conservative selection to the same peak response. A headroom CAViaR recursion and calibration wrappers follow existing methods (Engle & Manganelli, 2004; Gibbs & Candès, 2021). Calibration parameters update after outcomes mature. Fixed mean and median combinations use predefined members. Conservative selection favors the strongest shrinkage within one paired HAC standard error of the validation winner. This adapts one-standard-error selection (Hastie et al., 2009) to paired forecast losses.

Table 2: Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

Method / code	Forecast definition	Executed menu / setting
Historical headroom (F)	Current floor plus a quantile of completed peak headrooms	FX histories {720,2880}; weekly {52,260}
Translated path (S)	Translate completed signed paths to the current position before taking their peaks	Same completed-label pools; fixed-window and rolling-window controls
Historical peak (K)	Historical absolute-peak quantile projected to the current floor	Same histories and update controls
Quantile regression (QR)	Training-scaled headroom regression with ridge slopes and an unpenalized intercept	Original penalties $\{0.001, 0.01, 0.1\}$; expanded menu in Supplement
Local quantiles (KNN)	Neighbor headroom quantile, with a separate increment-scale variant	Training-only transforms and fixed neighbor menus; Supplement
Parsimonious regression (QR4)	Current signed/magnitude positions and short/long increment RMS	Separate five-point penalty menu; common-menu comparison in Supplement
Endpoint plug-in (L)	Floor plus a high absolute-displacement endpoint quantile	Proposition 2; empirical plug-in evaluated separately
Further benchmarks	Autoregressive replay; filtered paths; peak-tail fit; quantile forest	Existing bounded specifications, complete failures included in Supplement
Coverage and combinations	Delayed projected ACI-style wrappers; headroom SAV; fixed forecast combinations	Original-source wrapper/SAV inventory and broader combination inventory are separate

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in the Supplement; this table does not expand any executed menu.

6 Evaluation design

At refit r, training labels are eligible only when $i + H \leq r$. Predictors use issue-known observations, and training transforms remain fixed until the next refit. Original comparisons select parameters in the preceding validation segment. The full-prefix backcast holds its specified penalties fixed. Weekly selection uses 52 outer validation origins with overlapping labels.

Matched comparisons fix completed-response history and refit schedule for regression and historical headroom. The main FX reference uses a shorter rolling history. The main Treasury reference shares regression's longer history and refit schedule. Tables 3 and 4 report both alongside the other comparison sets.

Accuracy uses mean pinball loss; breaches are strict peak exceedances. Evaluation scores passive reports on the observed position paths. Paired loss differences use HAC covariance with the horizon-based bandwidth rule (Diebold & Mariano, 1995; Newey & West, 1987). Tables identify the small-sample adjustment and named multiplicity families (Harvey et al., 1997; Holm, 1979). Hit diagnostics use a fixed spaced grid and condition the transition likelihood on its initial hit (Kupiec, 1995; Christoffersen, 1998).

6.1 Research design and transparency

The original protocol fixes the two main comparisons and their reference procedures. Other p-values are exploratory and unadjusted unless stated. The Supplement records comparison families, execution chronology and complete failure inventories. The new stability screen and synthetic comparisons use a design fixed before their calculations. They retain separate estimation, simulation and reporting records.

7 Results

7.1 Where conditional models improve

Long-history FX regression beats matched historical headroom in both periods. Loss falls by 14.8% later and 16.6% earlier. Table 3 separates these matched controls from the shorter rolling reference.

Delayed calibration retains the later matched FX gain. Wrapped regression lowers loss by 15.0%. Its 329 breaches compare with 328 for wrapped historical headroom on common origins (Table 3).

The long-history FX gain persists across training-defined increment-RMS states and at the lower target quantile. At the lower target, matched loss falls by 10.1%, and loss against the original rolling reference by 9.2%. The Supplement reports all state results and failed validation at the upper quantile.

The full-prefix backcast also preserves reference-sensitive gains. Six-step regression lowers matched loss by 12.2% and rolling-reference loss by 5.3%. It records 416 breaches against 282 for matched headroom. The rolling-reference DM p-value is 0.055. Longer-horizon matched gains also occur, extending the pattern beyond the short-horizon configuration.

Table 3: Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

Period / procedure	Reference	Loss change (%)	Issues	Strict breaches (regression / reference)
Later / original main comparison	Shorter-history rolling headroom	-12.66	6625	376 / 332
Later / matched	Same-history fixed-window headroom	-14.77	6625	376 / 360
Earlier / original structure	Shorter-history rolling headroom	+3.17	1440	See complete earlier-period tables
Earlier / matched	Same-history fixed-window headroom	-16.62	1440	20 / 16
Later / coverage-wrapped	Same-history wrapped headroom	-15.00	6625	329 / 328

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

7.2 Where conditional models lose

The earlier FX ranking reverses when the reference changes. Against shorter-history rolling headroom, later loss falls by 12.7%, but earlier loss rises by 3.2%. Matched gains and rolling-reference gains therefore have different period stability (Table 3).

Short-history FX regression performs poorly at the longer horizon. Loss rises by 31.5% against matched historical headroom. Regression records 924 breaches against 427 for the reference; the full horizon map retains the other adverse comparisons.

Treasury-bill regression loses on both accuracy and breaches. Mean loss rises by 50.8%, with 54 regression breaches. Historical headroom records 28 breaches; the earlier short-window version also loses (Table 4).

Stronger regularization narrows weekly panel losses but leaves historical rules ahead. Median excess loss falls from 92.6% to 9.2% after menu expansion. Regression still loses 34 of 48 matched comparisons. Conservative selection leaves a worst-cell increase of 181.4% in the earlier weekly grid. This is the Treasury-bill short-window, 24-step comparison (Table 4).

Simpler conditional reports and fixed combinations also give mixed weekly results. The median combination lowers Treasury-bill loss by 4.0%, while the mean combination raises it by 2.8%. The headroom recursion is also worse. The Supplement retains every completed loss and model failure.

Historical headroom wins seven of eight same-record daily comparisons. The exception uses four predictors, history 260 and horizon 6, reducing loss by 3.0%. It records 37 breaches versus 30 for matched headroom. Daily aggregation therefore preserves the score–breach disagreement while changing the leading forecast in most comparisons.

7.3 Scores and breach behavior answer different questions

Lower FX loss coexists with more breaches before calibration. Regression breaches 376 times against 360 for matched historical headroom. Pinball loss measures error amounts, while a strict-hit indicator records event occurrence. Monitoring reports should show both.

Table 4: Weekly evidence across the original main comparison, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

Comparison set	Procedure	Loss change / median (%)	Higher-loss cells
Original Treasury-bill main comparison	Original regression, long history	+50.79	Original main comparison
Earlier Treasury-bill short-window version	Short-history short-window version	+144.30	Different history
Additional panel	Original full regression	+92.63	45 / 48
Additional panel	Four-predictor regression	+9.91	37 / 48
Additional panel	Full regression, expanded penalties	+9.22	34 / 48
Additional panel	Ordinary empirical headroom quantile	+0.50	30 / 48
Earlier weekly groups	Validation-minimum expanded rule	+4.15	26 groups
Earlier weekly groups	One-standard-error rule	+5.37	26 groups

The original bill main comparison has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results are a short-window version, not an exact replay of that main comparison. The one-standard-error worst excess is +181.39%; all its prewritten success criteria remain unmet. The Supplement lists all panel and selection results.

Elementary-score rankings cross for both original main comparisons (Figures 2 and 3). A separate 24-step FX comparison against rolling translated paths reverses near 20.6:1. Both six-step FX comparisons improve mean overrun and unused allowance. The weight reversal therefore belongs to the 24-step reference.

Loss and hit diagnostics answer different questions (Table 5). HAC tests compare average loss; strict-hit tests examine event frequency and serial behavior. RMS-state diagnostics compare conditional predictive ability across training-defined groups (Giacomini & White, 2006).

Table 5: Accuracy and hit diagnostics for explicitly named reports. Panel A reports paired working-reference tests; Panel B uses the fixed nonoverlap grid for hit tests. A better average score does not imply fewer breaches.

Panel A: paired contrast	DM p	HLN p	Named adjustment / p
FX original main comparison	2.20e-07	2.31e-07	Two-main comparison Bonferroni: 4.39e-07
FX matched history	4.47e-09	4.83e-09	Original matched Holm family: 2.86e-07
Treasury original main comparison	0.00761	0.00939	Original matched Holm family: 0.320

Original matched Holm uses 72 contrasts. The later regrouped matched family is separate; earlier-period writing-stage HAC comparisons have neither a new HLN field nor a new family adjustment.

Panel B: report	Grid hits	Grid issues	Kupiec p	Christoffersen CC p
FX regression	50	947	0.695	0.636
FX original rolling reference	49	947	0.807	0.905
FX matched fixed reference	49	947	0.807	0.646
Treasury regression	7	40	0.00413	0.0136
Treasury historical reference	4	40	0.199	0.262

Hit tests use the full fixed nonoverlap grid; the conditional-coverage likelihood conditions on its initial hit. Equality to the nominal tail rate is the stated diagnostic null, distinct from the population upper-quantile inequality. The grid does not create independence.

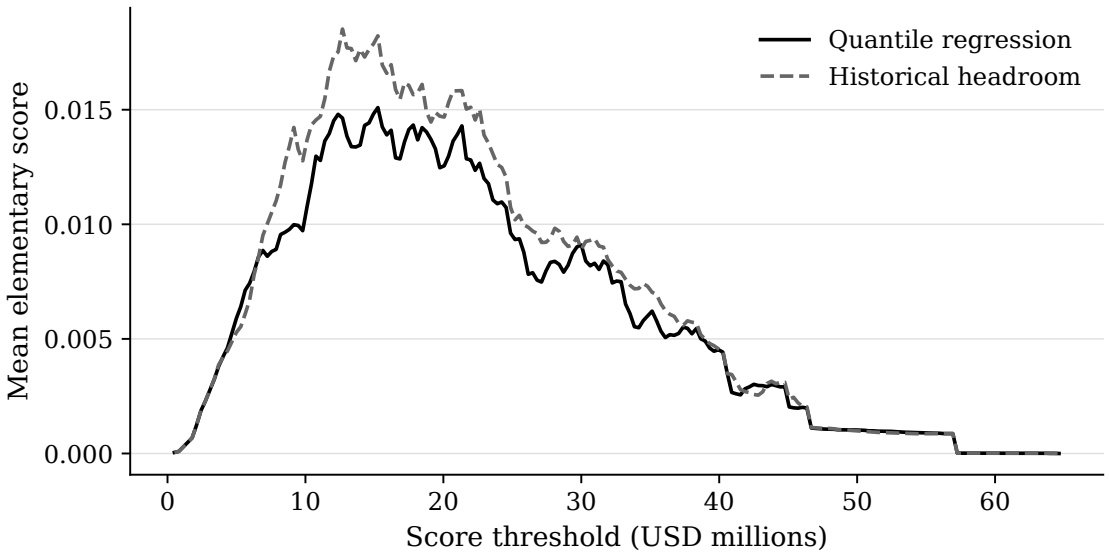


Figure 2: Murphy diagram for the main short-horizon FX comparison. The full-breakpoint calculation finds crossings; the graph is a display of saved scores. Average pinball superiority does not imply dominance across elementary scores.

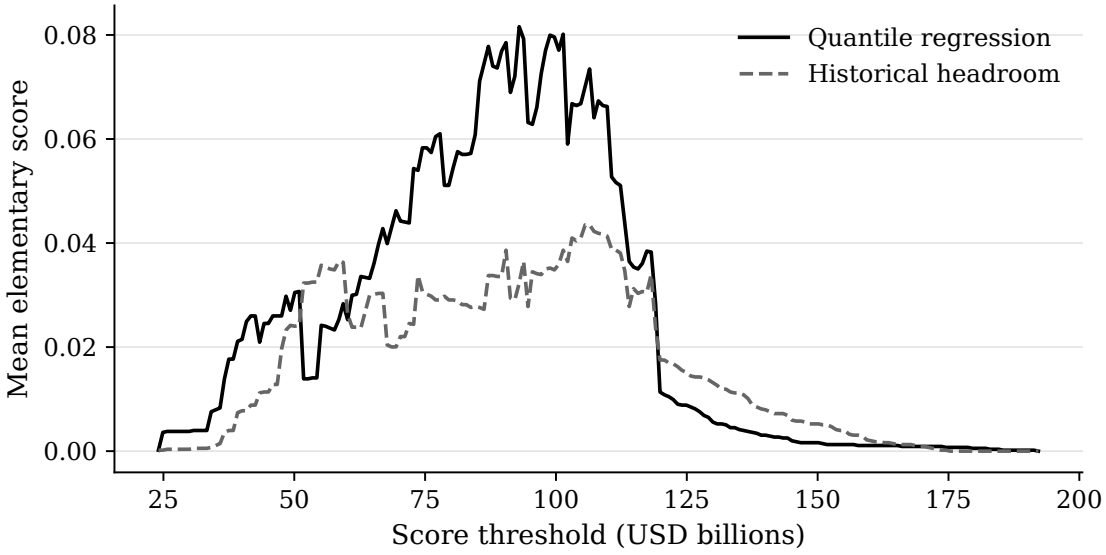


Figure 3: Murphy diagram for the main Treasury-bill comparison. The saved full-breakpoint comparison also crosses. A single average-score ordering leaves other consistent-score preferences unresolved.

7.4 Does sample size explain the low-frequency losses?

Short histories can produce large regression losses even under stationary laws. Loss magnitudes differ across designs with different menus, clocks, fitted laws and summaries. We retain each design's complete grid and distinguish its estimand from the others. Figure 4 summarizes the controlled and newly calibrated comparisons; complete settings appear in the Supplement.

The earlier weekly-calibrated design uses a six-candidate menu including an intercept-only limit. At history 52, its 33-predictor median changes are +1.1% under constant variance and +0.4% under weekly GARCH. At history 2880, they are -19.8% and -20.6%, respectively. Each complete cell uses 100 repetitions and 500 evaluation origins. Its control is constant-variance AR(1) levels, where the peak depends on the current position. Seven of its eight hourly cells fail numerical acceptance. Those whole cells remain unavailable. The prespecified constant-variance and short-weekly predictions are unsupported. Hourly failures leave the remaining mechanism predictions unassessable.

The controlled stationary grid uses narrower original menus and an hourly feature clock (Bollerslev, 1986). Its 33-predictor six-step loss increases by 113.5–135.5% at history 52 and 21.3–36.3% at history 260. These ranges cover nine settings at each history and use ratios of mean losses across repetitions. Each regression has clear gains in 2 of 12 targeted cells and 6 of 60 outside cells. The complete 72-cell grid therefore contradicts the proposed narrow gain region. It also shows that stationary short-history losses were already present before the new experiment.

The new experiment calibrates constant and segmented AR–GARCH–t processes to pre-evaluation prefixes of thirteen weekly series. The joint OLS score uses Hansen's statistic (Hansen, 1992). A fitted constant-parameter null supplies its parametric bootstrap reference. Seven series reject at the pointwise 5% level; Holm adjustment retains two. A constrained descriptive partition selects up to two changes. The switching arm cycles the fitted regimes; its constant counterpart retains the whole-prefix parameters. The arms transform common uniforms through their own innovation laws and use the controlled grid's forecasting procedures.

At history 52, the across-source median loss increase is 116.2% for the constant arm and 121.4% for switching. These are equal-weight medians of thirteen source medians, each based on 32 paired repetitions. The constant result misses the preregistered near-parity prediction. At his-

tory 260, one source's required 33-predictor repetition fails in each arm. Consequently, both full thirteen-source summaries are unavailable, and the two-history switching prediction is unassessable. Neither the successful repetitions nor the remaining sources replace those missing complete cells.

The daily fitted null fails the fourth-moment qualification, and the matched-calendar hourly null fails optimization. Their requested diagnostic comparison is also unassessable. The observed weekly rejection count supports the registered screening threshold, while the remaining results leave the proposed explanation unresolved. Instability is not necessary for large losses in these simulated procedures: the fitted constant controls already generate them. The evidence leaves the empirical low-frequency mechanism unidentified.

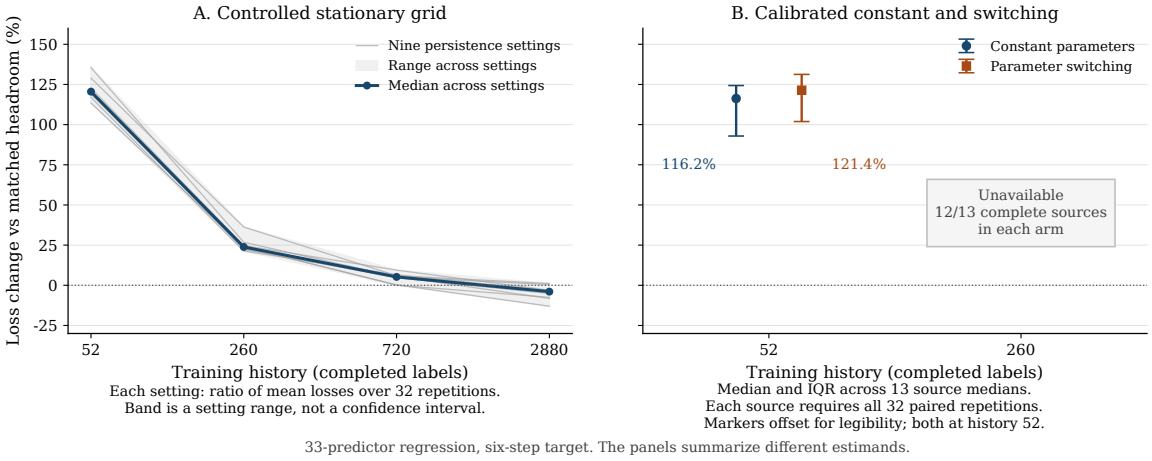


Figure 4: Stationary short-history losses and the calibrated switching comparison. Left: relative mean-loss changes for nine controlled six-step settings, with their median and full range. Right: medians across thirteen source medians with source IQR bars, offset horizontally at history 52 for visibility; full history-260 summaries remain unavailable.

8 Implications for model validation

Benchmark against historical headroom. Use the same completed labels, history length and refit schedule. Name the shorter rolling reference separately, since its FX ranking changes across periods.

For these short weekly samples, use corrected-rank history as the default benchmark. Expanded penalties and conservative selection leave large losses. The ordinary empirical quantile is up to 36.5% worse than corrected headroom in the earlier weekly grid.

Report breaches with the score. Show mean underprediction, mean unused headroom and strict-breach count on the same grid. The unwrapped matched FX gain records 376 versus 360 breaches; the daily regression exception also has more breaches. Table 6 organizes these validation questions.

Table 6: Evidence map for model validation. The daily and new-study columns locate supporting comparisons for each setting. A dash marks a task outside that comparison's scope.

Frequency / history	Horizon	Financial evidence	Same-record daily contrast	New stability study	Validation action
Hourly / longer history	Short	Repeated matched gains; delayed calibration retains the later gain	—	—	Match the reference history and report breaches
Hourly / shorter history	Short	Small score gain with more breaches	—	—	Require stable joint performance across periods
Hourly / either history	Longer	Matched backcast gains coexist with reference-sensitive losses; horizon crossings remain	—	—	Check reference, horizon order and error preferences
Daily / same dealer	Observed days	Historical score wins predominate	Historical headroom wins 7/8; the four-predictor exception has more breaches	Daily moment qualification and hourly optimization fail; the diagnostic comparison is unavailable	Retain historical controls and assess their own breaches
Weekly / short or long history	Short or longer	Regression often loses despite expanded penalties	—	At history 52, full-regression medians are +116.2% with constant parameters and +121.4% with switching; full history-260 medians are unavailable	Use corrected-rank history as the default benchmark for these short samples; compare scores and breaches
Any source / endpoint bound	Specified horizon	Population premises and empirical plug-in behavior differ	—	—	Check absolute displacement and dependence

9 Limitations

The FX evidence describes a single market maker's client-flow record. Its original evaluation period was viewed before the conditional methods were specified. Rules for the earlier 2015 comparison followed the original results. The full-prefix backcast uses penalties selected after its issue dates and contains the earlier backcast. Its target paths were covered by previous scoring. Both exercises remain retrospective.

The weekly categories share a reporting system and economic components; they are related aggregate tasks, not independent dealer replications. Treasury comparisons replay the downloaded economic-date vintage. Historical publication-time information is not reconstructed. Aggregation can offset gross dealer exposure; sampled net inventory also differs from settlement cash needs. Valuation and reporting-composition changes are not separated from trading. Sampled peaks omit intervening excursions. Daily last records also differ from verified closes or intraday maxima; the source timezone and release vintage are unknown.

Daily aggregation jointly changes physical histories, horizons, update cadence, calendar predictors, evaluation span and sampled targets. The same-record comparison leaves a pure frequency effect unidentified.

The design gives no causal interpretation or individual-predictor effects. The financial procedures provide no coverage guarantee under their dependence and calibration scheme. Trading and funding gains are not measured. Scoring weights are evaluation preferences, not estimated intervention or funding costs. Independent dealer records and prospective observations would broaden validation; finer exposure and cost data would connect reports to inventory actions.

Short overlapping validation samples and research-stage selection limit tuning stability and confirmatory inference. Longer weekly outer validation windows remain untested. The simulation designs use different menus, clocks, calibration laws and summary statistics. Their differences cannot isolate a single source of the empirical losses. The new parameter-switching process repeats fitted dwell lengths and adds artificial last-to-first transitions. Its stability screen uses a plug-in model-based approximation. Moment qualification and successful optimization leave the assumed law and boundary-estimate uncertainty unresolved. Failed cells and unavailable diagnostic comparisons retain those statuses.

10 Conclusion

Simple historical rules are hard to beat for peak net positions. Hourly FX regression lowers matched loss by 14.8% and 16.6% across periods. Delayed calibration retains the later gain with 329 versus 328 breaches. Rankings against the shorter rolling reference change by period.

Weekly regression still loses 34 of 48 expanded-panel comparisons, and historical headroom wins most same-record daily comparisons. Conservative validation leaves substantial losses. Stationary controlled and fitted constant-parameter processes also produce large short-history losses. At history 52, the newly fitted constant control yields a median increase of 116.2%, compared with 121.4% under switching. The low-frequency mechanism remains unidentified. Peak-limit validation should compare accuracy, error amounts, breaches and reference stability together.

Data and code availability

Data availability. The FX input is the specified version of the Kaggle Net Position Time Series Dataset. The owning metadata labels its license "Data files © Original Authors." This repository supplies the acquisition route and input fingerprint, but excludes raw FX observations and private per-issue prediction vectors. Obtain the source from `https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset` subject to the original rights holders' terms.

The New York Fed aggregates are available from its Primary Dealer Statistics service under the Bank's Terms of Use. The repository preserves the unchanged accepted Treasury snapshot with attribution and identifies study calculations separately. Copying or distributing that content remains subject to its source notices, attribution, modification labelling, continuing conditions and endorsement restrictions: `https://www.newyorkfed.org/privacy/termsofuse.html`. The code license does not replace those source conditions.

Code availability. Code, saved results and version-specific reproduction instructions are available at `https://github.com/ByronLi0207/financial-mathematics-paper-20261020/tree/revision-v9`. The package includes all earlier comparisons, calibrated simulations and the new stability study. Document compilation and restricted-input financial replay have separate instructions. Complete, adverse, failed and unavailable outcomes are retained.

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AI statement: [TO BE COMPLETED BY AUTHOR].

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